ECE 555 Project Schematics

Fall 2026

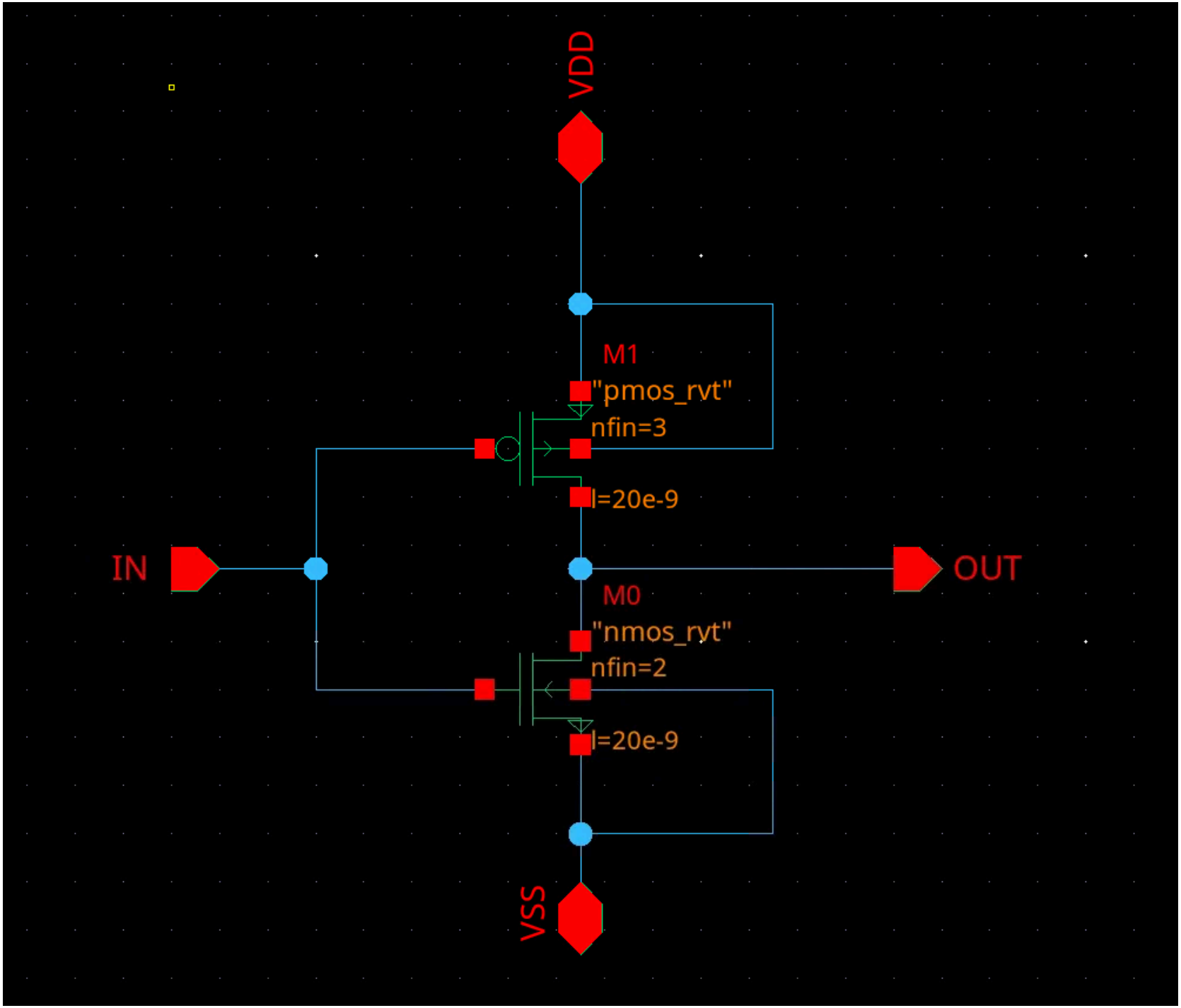
Team

Jaeseong Lee

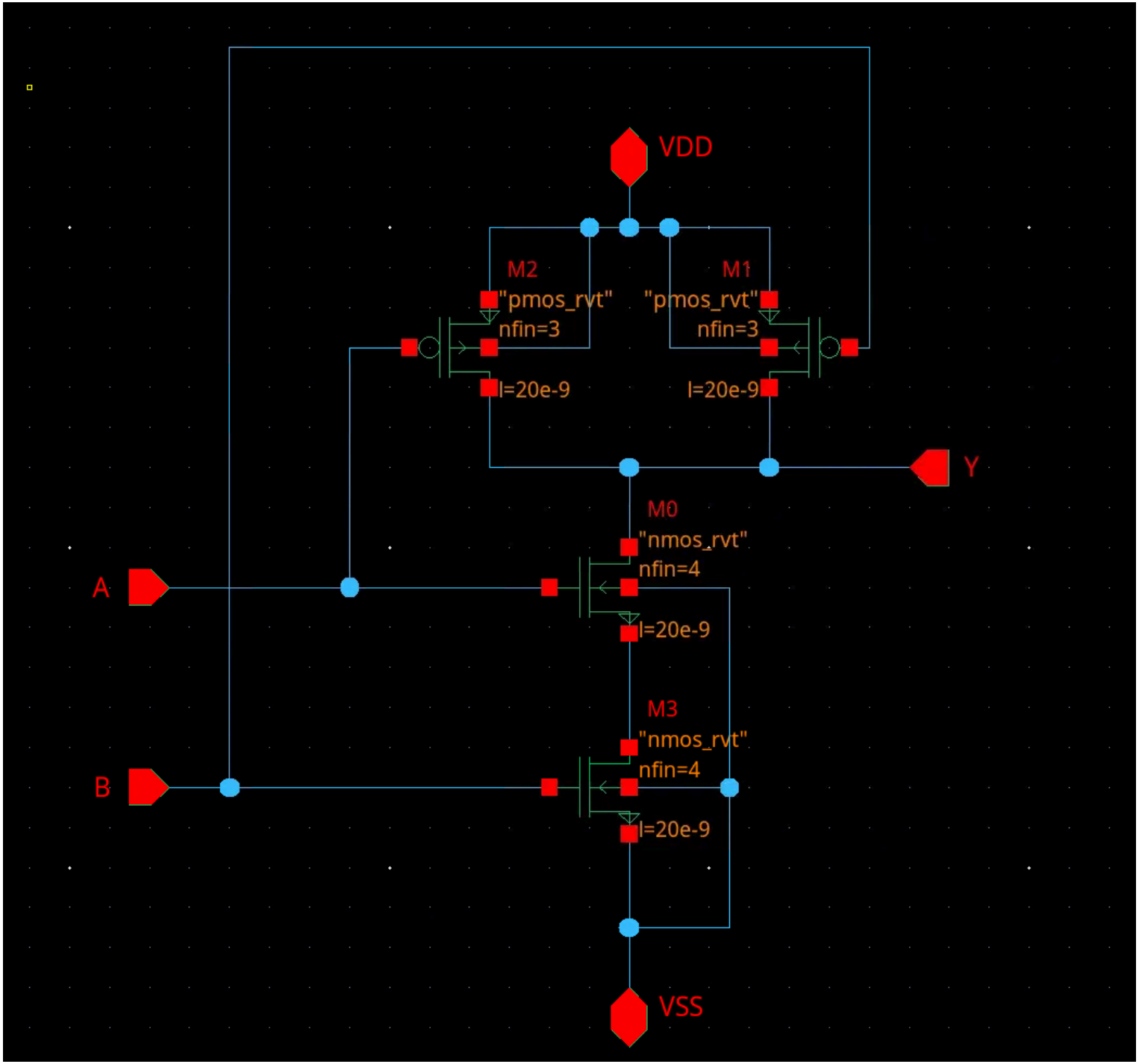
Josh Amann

Karol Yan

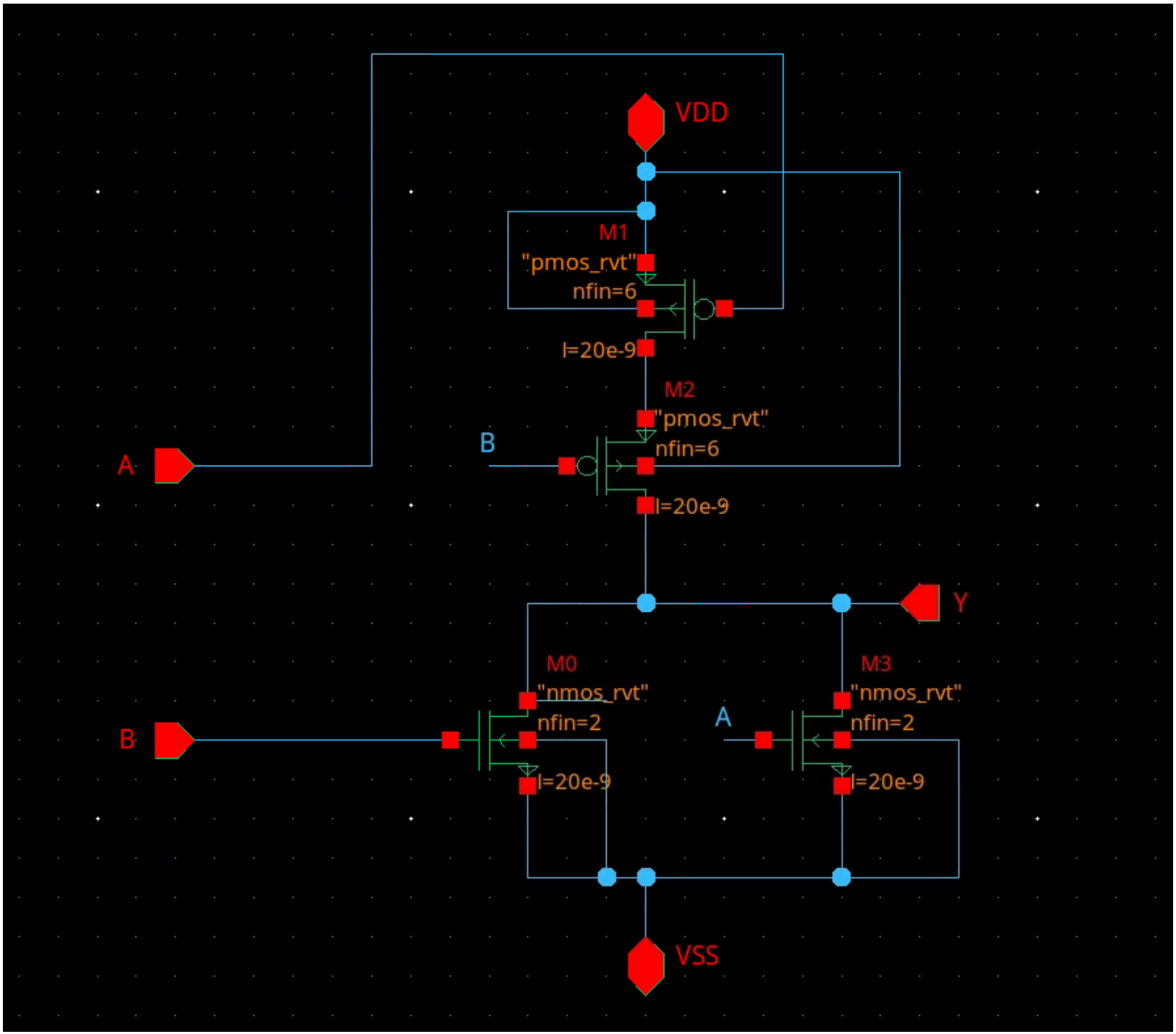
1. Inv



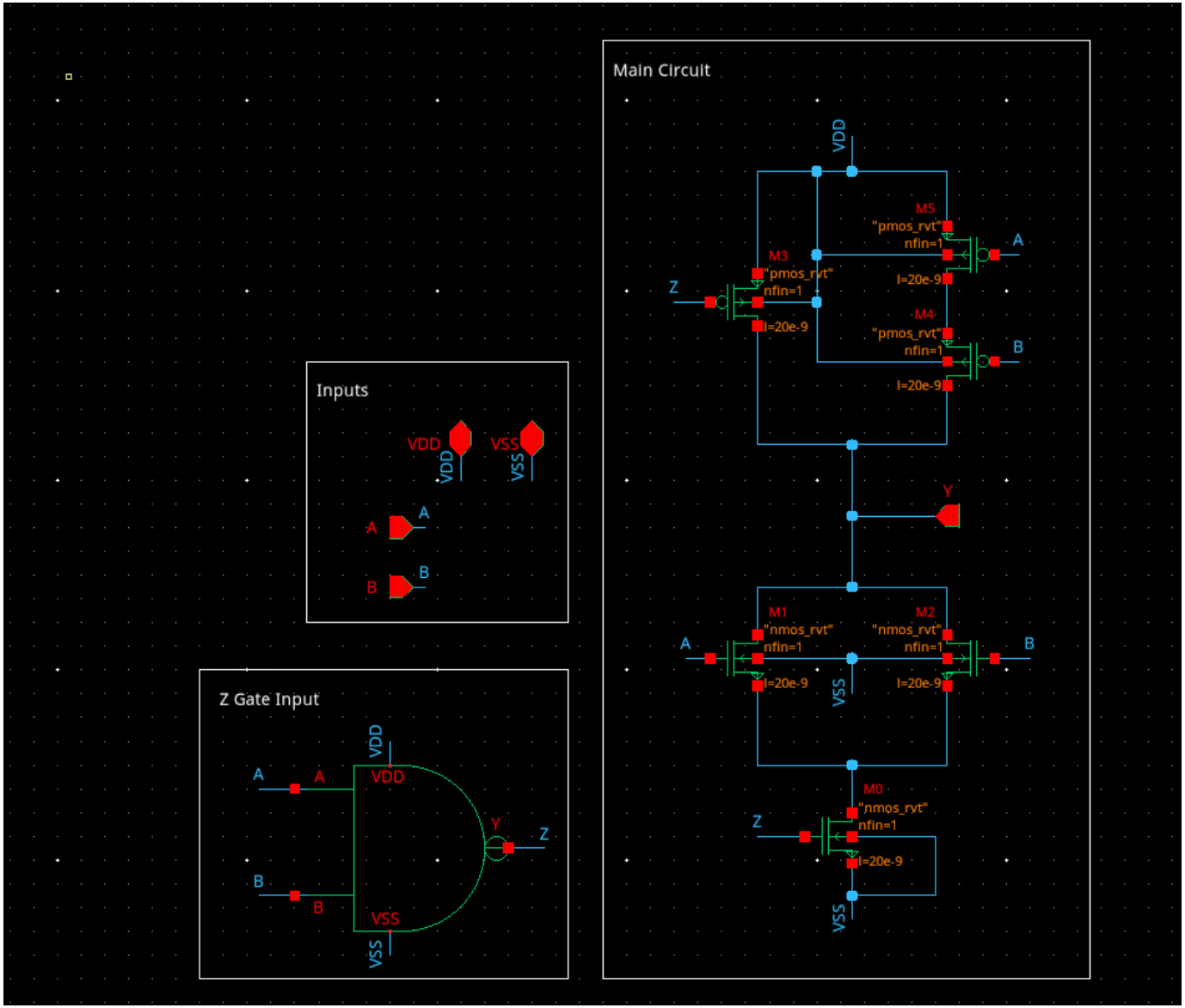
2. NAND2


VDD
M2
"pmos_rvt"
nfin=3
l=20e-9
M1
"pmos_rvt"
nfin=3
l=20e-9
Y
M0
"nmos_rvt"
nfin=4
l=20e-9
A
M3
"nmos_rvt"
nfin=4
l=20e-9
B
VSS

3. NOR2

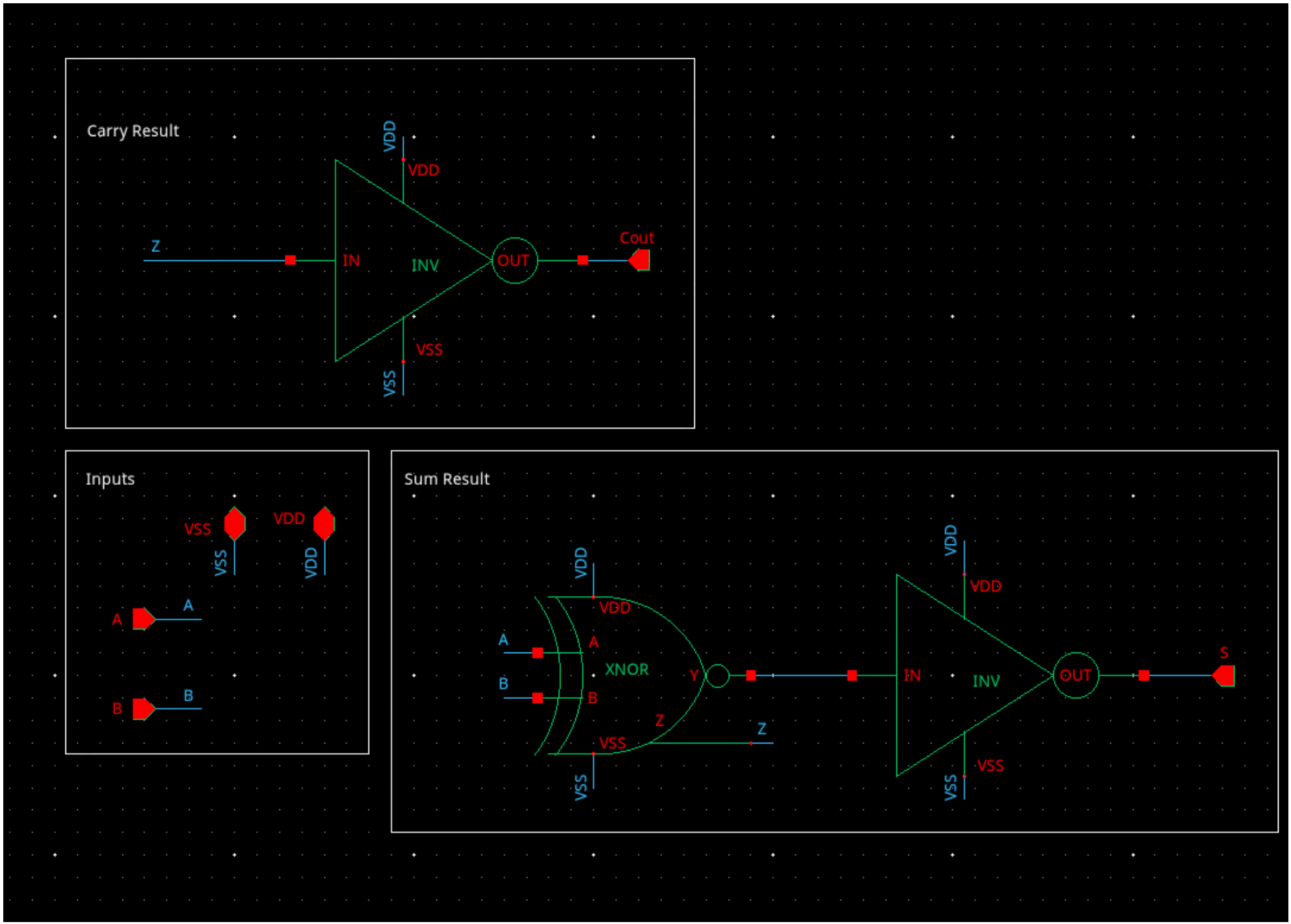

VDD
M1
"pmos_rvt"
nfin=6
l=20e-9
M2
"pmos_rvt"
nfin=6
l=20e-9
A
B
Y
M0
"nmos_rvt"
nfin=2
l=20e-9
M3
"nmos_rvt"
nfin=2
l=20e-9
B
A
VSS

4. XNOR

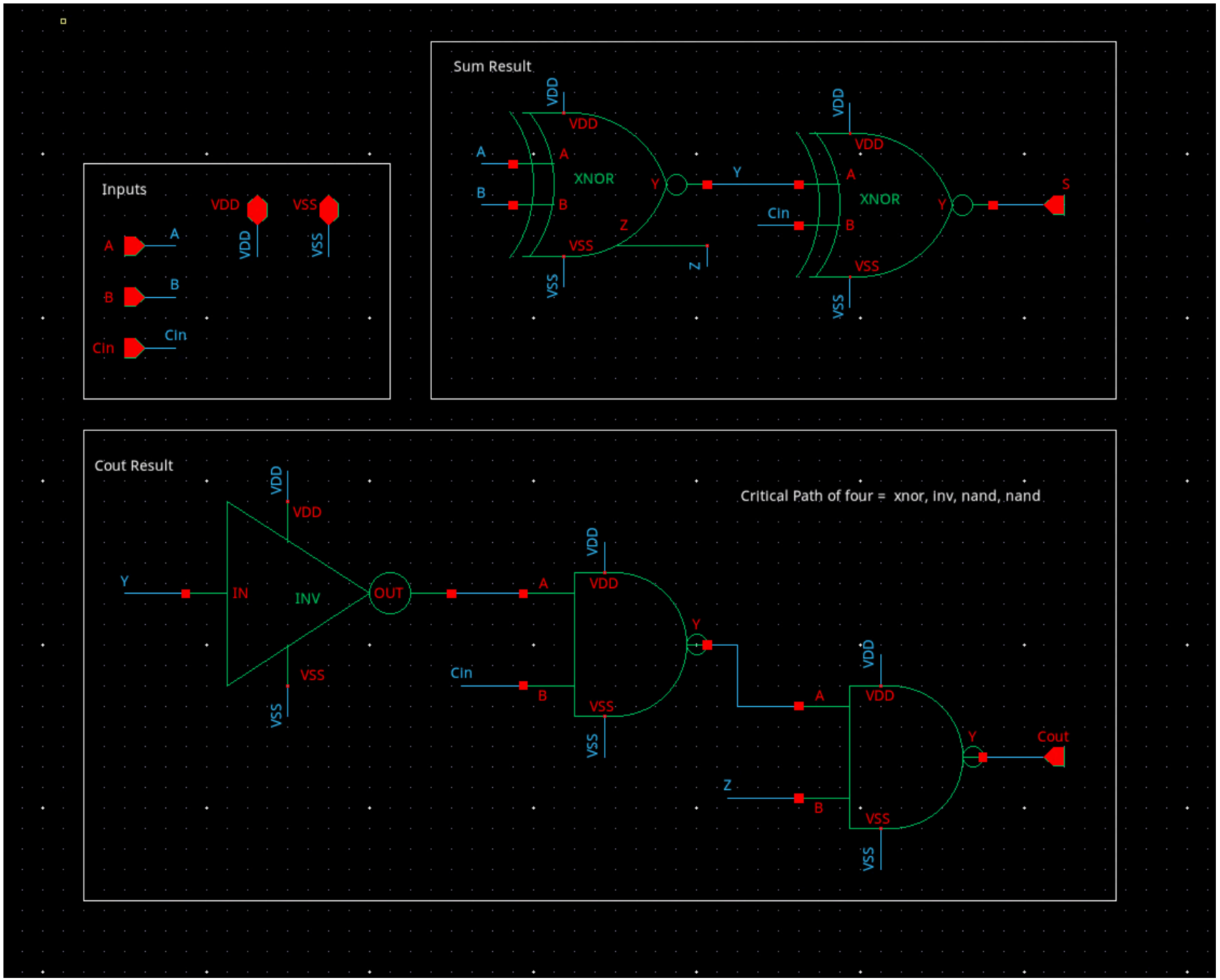


Note: Transistor sizing (beta ratio) is not correct yet. All devices are currently nfin=1 and still need to be sized to balance the pull-up and pull-down networks.
Note: Z (= NAND(A,B), the internal 'Z Gate Input') is also brought out as an output pin. In some instances Z is used as an output - e.g. the Half Adder uses it for Cout and the Full Adder uses it in the carry logic. Other instances (e.g. the Multiplier) leave it unused.

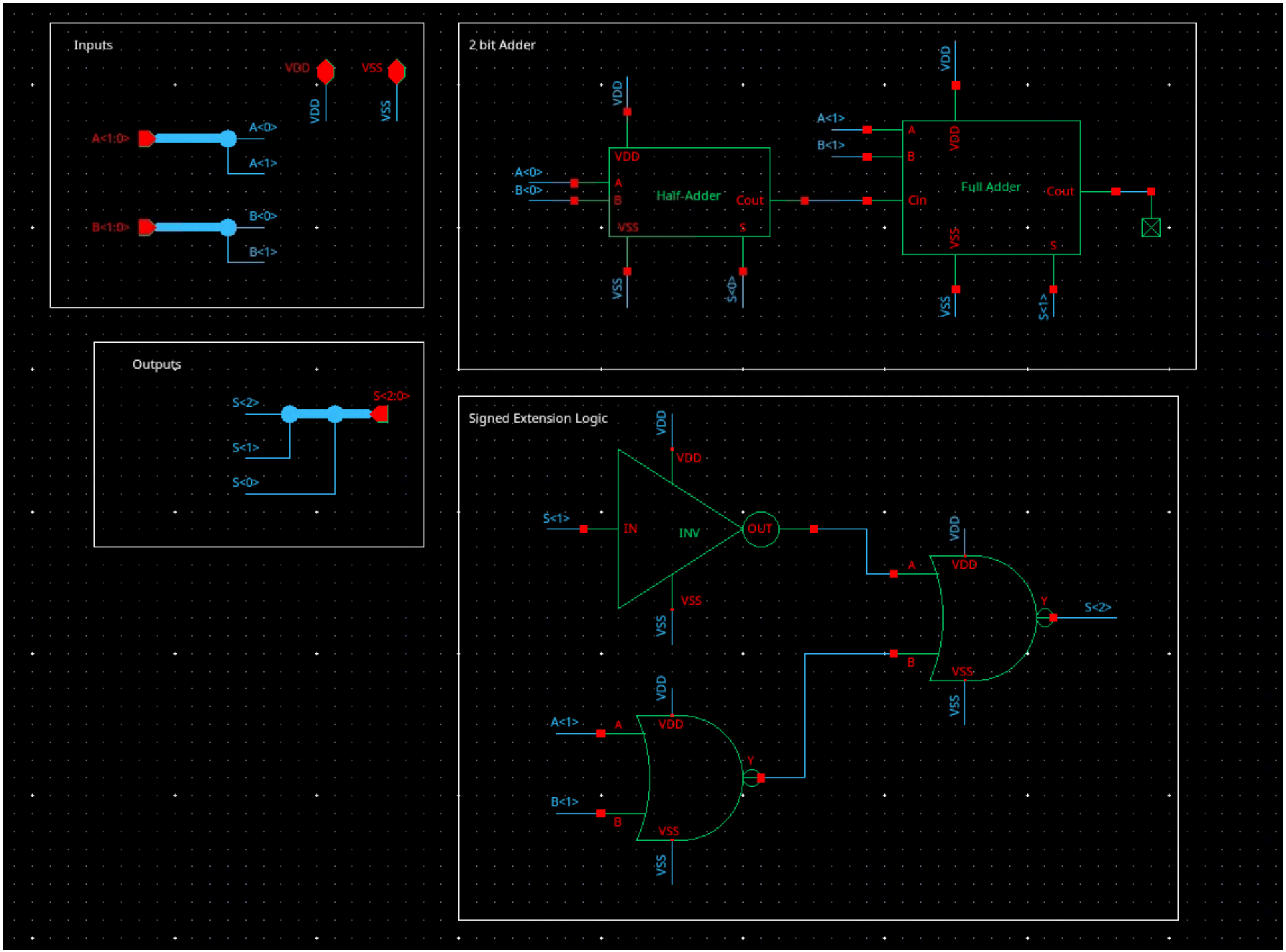
5. Half Adder


Carry Result
VDD
VDD
Z
IN
INV
OUT
Cout
VSS
VSS
Inputs
VSS
VSS
VDD
VDD
A
A
B
B
Sum Result
VDD
VDD
A
A
XNOR
Y
B
B
Z
Z
VSS
VSS
VDD
VDD
IN
INV
OUT
S
VSS
VSS

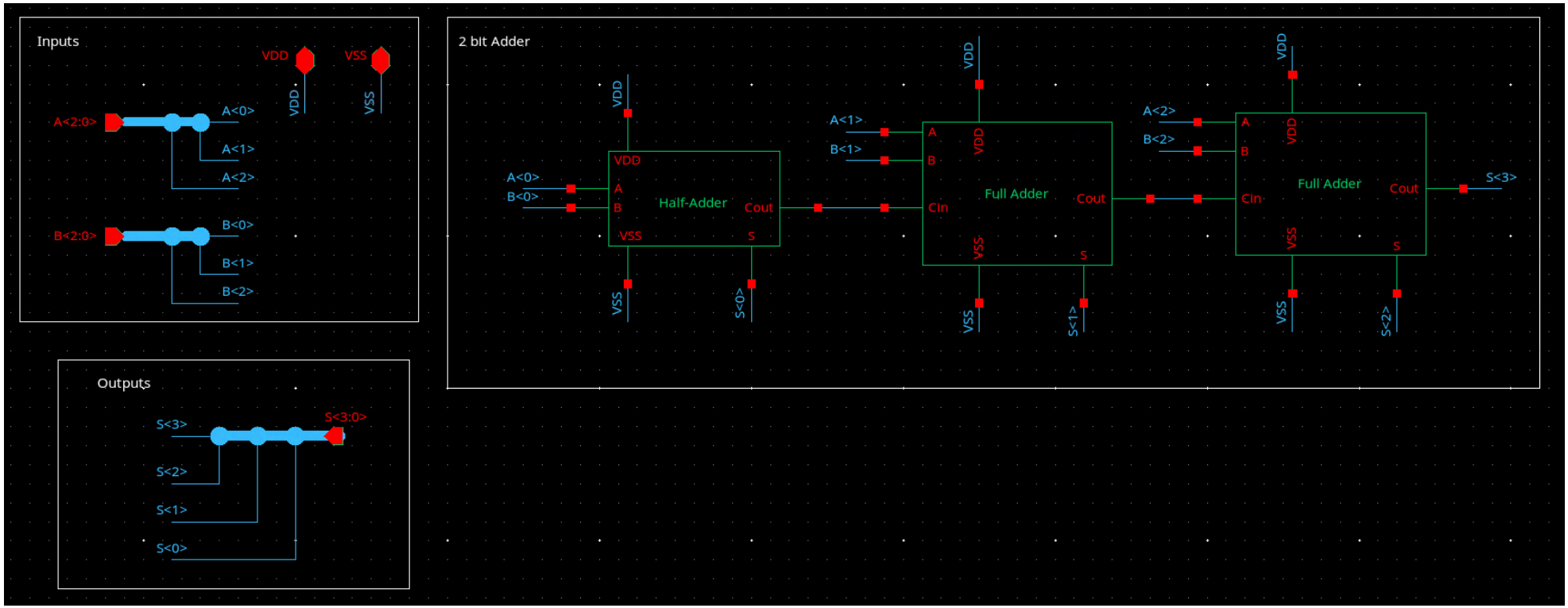
6. Full Adder


Inputs
A
B
Cin
VDD
VSS
Sum Result
XNOR
XNOR
Y
Z
S
Cout Result
Critical Path of four = xnor, inv, nand, nand
IN
INV
OUT
Cout

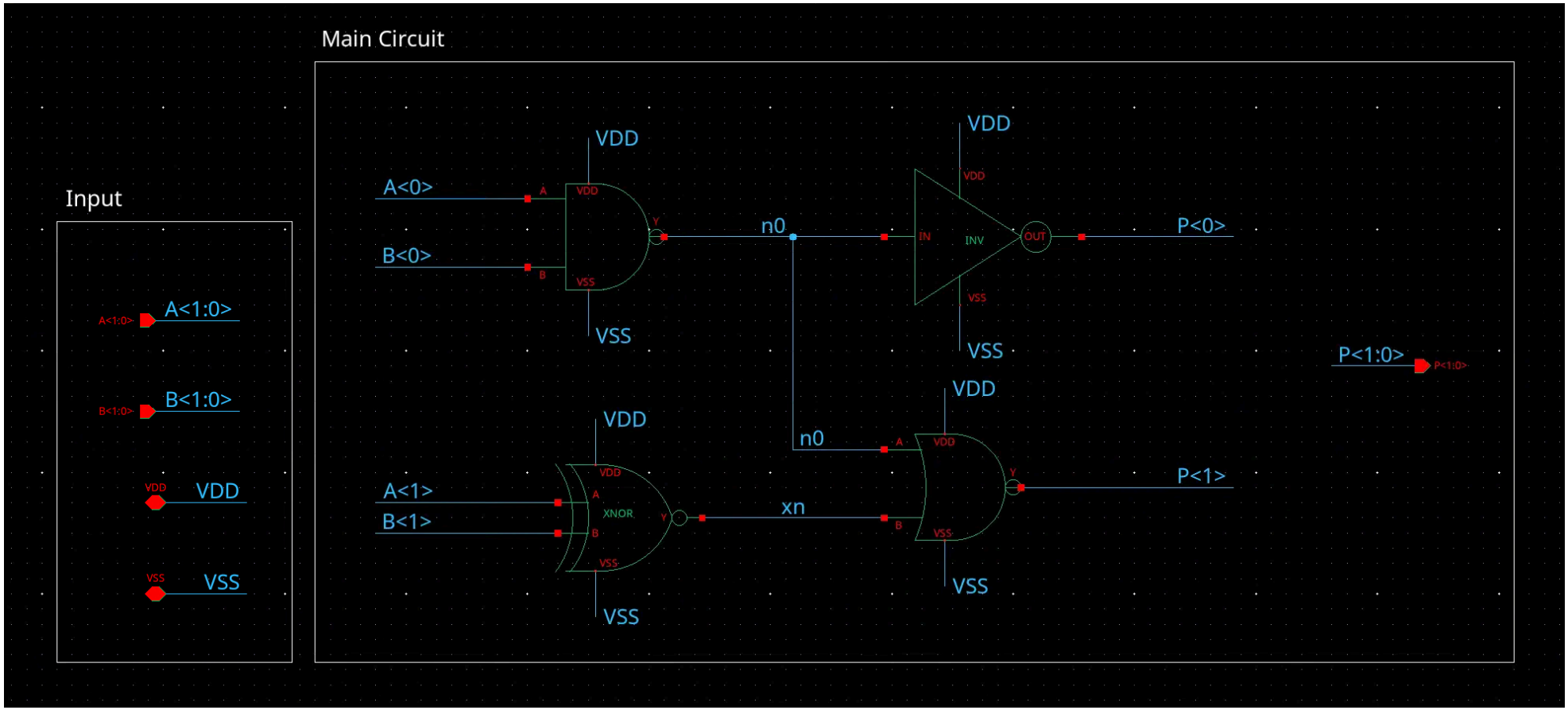
7. 2 bit Adder


Inputs
VDD
VSS
A<1:0>
A<0>
A<1>
B<1:0>
B<0>
B<1>
Outputs
S<2:0>
S<2>
S<1>
S<0>
2 bit Adder
Half-Adder
Full Adder
Cout
Cin
Signed Extension Logic
IN
INV
OUT
Y

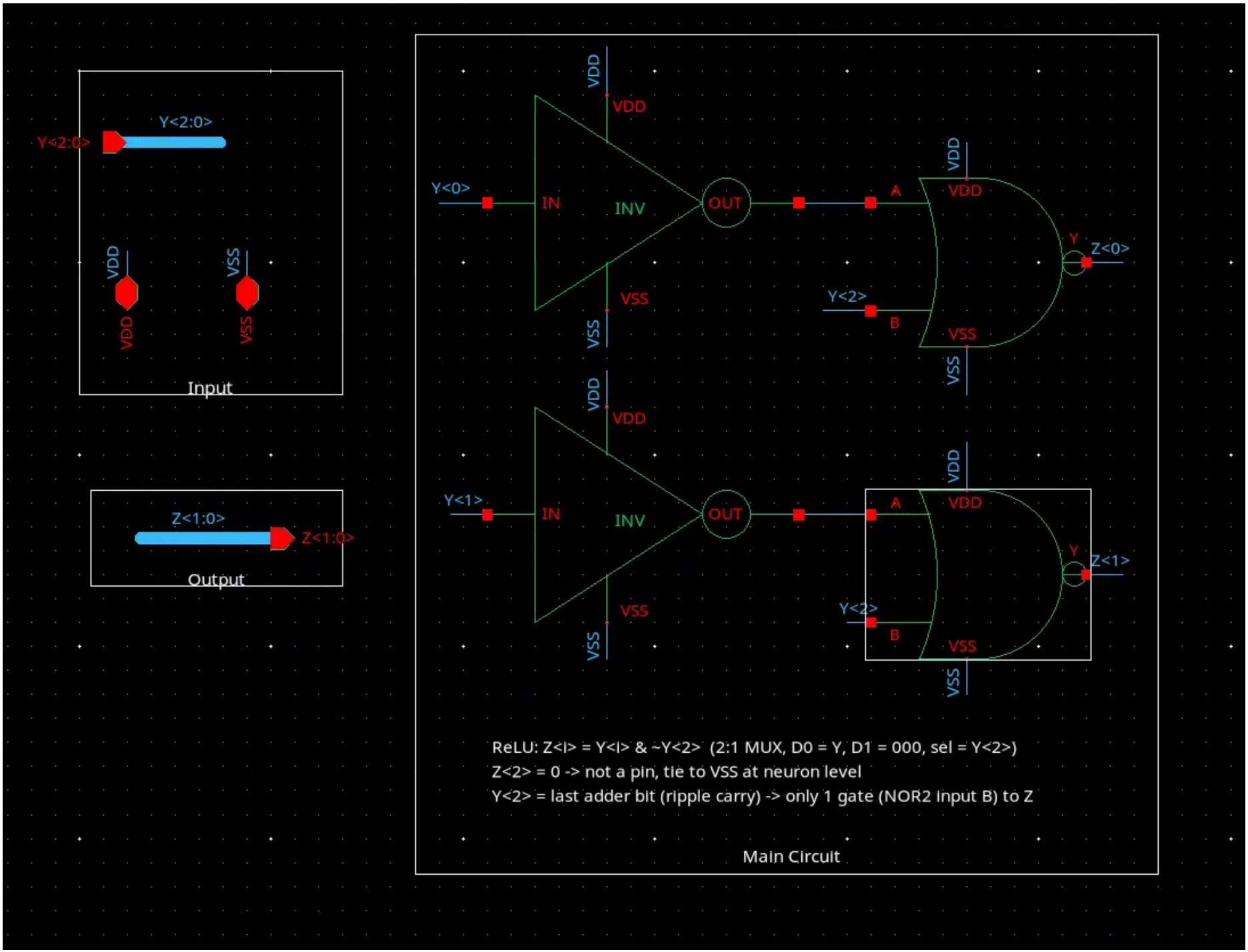
8. 3 bit Adder


Inputs
VDD
VSS
A<2:0>
A<0>
A<1>
A<2>
B<2:0>
B<0>
B<1>
B<2>
Outputs
S<3:0>
S<3>
S<2>
S<1>
S<0>
2 bit Adder
Half-Adder
Full Adder
Full Adder
Cout
Cin
S<3>
S<2>
S<1>
S<0>

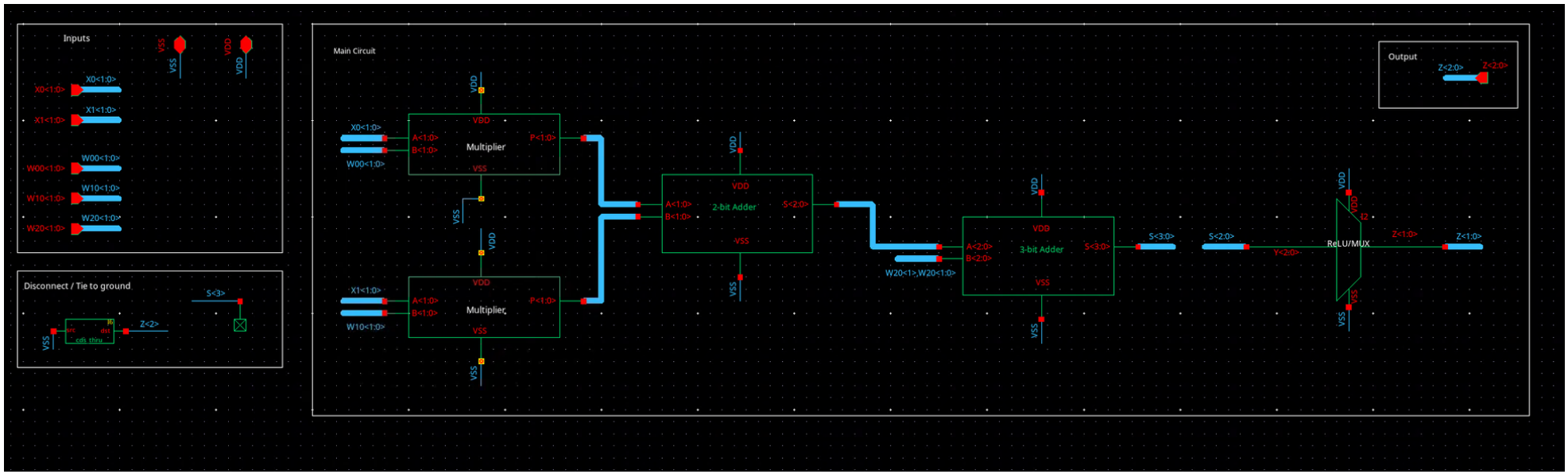
9. Multiplier



10. MUX


Y<2:0>
Y<2:0>
VDD
VDD
VSS
VSS
Input
Z<1:0>
Z<1:0>
Output
VDD
VDD
Y<0>
IN
INV
OUT
VSS
VSS
A
VDD
VDD
Y
Z<0>
Y<2>
B
VSS
VSS
VDD
VDD
Y<1>
IN
INV
OUT
VSS
VSS
A
VDD
VDD
Y
Z<1>
Y<2>
B
VSS
VSS
ReLU: Z<I> = Y<I> & ~Y<2> (2:1 MUX, D0 = Y, D1 = 000, sel = Y<2>)
Z<2> = 0 -> not a pin, tie to VSS at neuron level
Y<2> = last adder bit (ripple carry) -> only 1 gate (NOR2 input B) to Z
Main Circuit

11. Neuron


Inputs
VSS
VDD
X0<1:0>
X1<1:0>
W00<1:0>
W10<1:0>
W20<1:0>
Disconnect / Tie to ground
S<3>
Z<2>
Main Circuit
Multiplier
A<1:0>
B<1:0>
P<1:0>
Multiplier
2-bit Adder
S<2:0>
3-bit Adder
A<2:0>
B<2:0>
S<3:0>
W20<1>,W20<1:0>
Output
Z<2:0>
Y<2:0>
ReLU/MUX
Z<1:0>
t2